A quadratic equation with no integer solution
Ireland–Rosen, 3.4
Show that the equation $3x^2 + 2 = y^2$ has no solution in integers.

Proof. Suppose, for a contradiction, that x and y are integers satisfying

$$3x^2 + 2 = y^2.$$

Consider y modulo 3. Its remainder is one of 0, 1, or 2:

$$y \bmod 3 = 0 \quad \text{or} \quad y \bmod 3 = 1 \quad \text{or} \quad y \bmod 3 = 2.$$

Since $y^2 = y \cdot y$, reducing modulo 3 gives

$$y^2 \bmod 3 = \big((y \bmod 3)(y \bmod 3)\big) \bmod 3.$$

We check the three cases:

- If $y \bmod 3 = 0$, then $y^2 \bmod 3 = (0 \cdot 0) \bmod 3 = 0$.
- If $y \bmod 3 = 1$, then $y^2 \bmod 3 = (1 \cdot 1) \bmod 3 = 1$.
- If $y \bmod 3 = 2$, then $y^2 \bmod 3 = (2 \cdot 2) \bmod 3 = 4 \bmod 3 = 1$.

Thus in all cases,

$$y^2 \bmod 3 = 0 \quad \text{or} \quad y^2 \bmod 3 = 1.$$

On the other hand, from the assumed equation $y^2 = 3x^2 + 2$, reducing modulo 3 gives

$$y^2 \bmod 3 = (3x^2 + 2) \bmod 3 = 2,$$

since $3x^2$ is divisible by 3. This contradicts $y^2 \bmod 3 = 0$ or $y^2 \bmod 3 = 1$.

Therefore $3x^2 + 2 = y^2$ has no solution in integers. □